SOLID WASTE MANAGEMENT MADE EASY











A Do-it Yourself Guide to
a Community-Based Ecological
Solid Waste Management Programme



Solid Waste Management Made Easy

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Message



Since its enactment in 2000, Republic Act 9003, or the Ecological Solid Waste Management Act, has provided the framework to guide the Philippines towards a waste-free society. Through this legislation, the government is able to institutionalize a national program that now manages the "control, transfer, transport, process, and disposal" of solid wastes in the country.

For the past ten years, local government units have been actively pursuing the mandates of RA 9003 – segregation at source, segregated collection, recycling, and resource recovery, mandatory establishment of Materials Recovery Facilities (MRFs) in every *barangay* or cluster of *barangays*, the banning of open burning, and the closure of open dumps and controlled disposal facilities. Environmental drives and campaigns have also been organized to make Filipinos more knowledgeable of the benefits of proper waste management, how it can help not just the environment but also improve the health and state of living of our people.

Together with our friends from the non-government organizations and private corporations, campaigns and drives have been pushed. We organized the very successful Nationwide Search for Model Barangays on Ecological Solid Waste Management to which we discovered that at the heart of our regions are provinces that are exemplary in their waste management practice.

It is with pride that we recognize the many initiatives of our local government units as truly commendable – from clustering to the establishment of Materials Recovery Facilities; closure and rehabilitation of dumpsites into ecological parks; the establishment of ecological solid waste management system; massive information and education campaigns, trash to cash projects; and the development of alternative technologies to process biodegradable, recyclable, and residual wastes. Through the commitment and tireless dedication of our local leaders, we are now witnessing communities become, not just aware, but more importantly, involved in our campaign towards cleaner cities and municipalities.

All these positive development, we attribute to the heightened consciousness that every individual, every member of the community, plays a significant role in transforming and rehabilitating our society.

In 2007, through the generous support of the United Nations Development Programme, we have produced this guidebook, "Solid Waste Management Made Easy: A Do-It-Yourself Guide to a Community-Based Ecological Solid Waste Management Programme". One of the most sought-after reference materials on proper community waste management, we hope that this reprint would serve the needs of even more cities and municipalities who are now developing their own ecological solid waste management programs.

Sa sama-samang pagkilos, basura ay maisasaayos. Let us work together to improve our environment.

SECRETARY RAMON J.P. PAJE
Department of Environment and Natural Resources
and Chairman, National Solid Waste Management Commission

Message



From the ten Metro Manila communities piloted by the United Nations Development Programme to be models on ecological solid waste management under their Community-Based Ecological Solid Waste Management Programme in the Philippines, we now have hundreds of local government units developing and sustaining their respective waste management systems.

The national government and the local government units have merged, along with the private sector and the non-government organizations to bring us to the Philippines we have now – a country with people working together, placing extreme importance on partnerships, to pave the way for a cleaner environment.

The National Solid Waste Management Commission-Secretariat, for its part, has been promoting technologies that would process wastes into useful products, as well as keeping our focus on the proper closure and rehabilitation of the open and controlled dumpsites nationwide. We advocate the development of clustered sanitary landfills and other cleaner and safer forms of disposal system, and we continuously provide guidance and technical assistance on concerns on waste management strategies and options to our LGUs. We believe and we push for the promotion of ecological solid waste management through capacity building, advocacy, and information and education campaigns in the country.

Last year, the Department of Environment and Natural Resources, through its Environmental Management Bureau, intensified the campaign to clear our esteros and water bodies of solid wastes. Dubbed as the "Adopt-an-Estero Project", private corporations provide the much-needed assistance to the local governments, particularly the *barangays*, to ensure that the wastes that have been clogging our esteros and polluting our water bodies will be trapped and excavated, and that once the water bodies have been rehabilitated, would be maintained by the residents of the community. Under this program, the Commission serves an important role of providing technical assistance in establishing the appropriate Materials Recovery Facility/System, as well as information and education support for all concerned *barangays*.

Solid wastes may be the country's worst environmental problem but we maintain, and we all must recognize that its management is the simplest, and perhaps, the single most important thing we could do for our environment. If we unite and practice proper ecological solid waste management, and if all could just get involved, a margin of the Philippines' climate change, air and water pollution problem will be solved as well.

This guidebook, "Solid Waste Management Made Easy", provides you with practical solutions and step-by-step guide on how solid waste management is done. Success stories of community-based ecological solid waste management are also shared to even inspire you more on how truly easy it is to manage wastes. We sincerely hope that through this book, you will be part of the millions of Filipinos who have changed their ways for the better, through the very simple act of proper solid waste management.

MS. EMELITA C. AGUINALDO
Executive Director
National Solid Waste Management Commission

By the Numbers

½ kilo

the average trash generated per person per day in Metro Manila, according to the ADB. This translates to 7,000 tons of trash per day.

Php 3.8 billion

the annual total spending of Metro Manila for garbage collection. This translates to Php1,500 per ton of garbage or Php10.5 million per day.

Php 3.61 billion

will be saved if recyclables and biodegradables are segregated at source and do not end up in dumpsites. This translates to cost savings of 95%.

(Source: ADB study, 2003)

WHAT P3.61 BILLION CAN BUY



7,220
classrooms (assuming one classroom is Php500,000) to benefit 433,200 pupils (if one classroom holds 60 pupils), ***or***

20,056
additional teachers (monthly pay of Php15,000 per teacher), ***or***

20,056
additional policemen (Php15,000 per policeman monthly), ***or***

361
kilometers of roads (at Php10 million per kilometer), ***or***

7,220
additional low-cost houses (Php500,000 per two-bedroom house)

At least 200

people who died after an avalanche of garbage collapsed in the Payatas dumpsite in July 2000. *(Source: ADB Review, 2000)*

25%

target percentage reduction of waste from 2002 to 2010 under the UNDP Community-Based Ecological Solid Waste Management Programme.

What this handbook is about

THANK YOU for picking up this handbook and opening its first few pages. As the title goes, this do-it-yourself guide to handling solid waste (*basura)* is designed to make us appreciate the intent of the law (Republic Act 9003, the Ecological Solid Waste Management Act of 2000) and help us divert 95% of our waste from the dumpsite or landfill. We have also published a fieldbook companion to this handbook that contains training modules on running your own ecological solid waste management program.

Solid waste management is not just the responsibility of the government or our top local leaders. We are all in a position to make a difference in our barangays or communities. By taking interest in what the handbook has to say, we could be making the first step in saving, not just the environment, but a lot of people's lives in our community as well.

Managing our solid waste is no longer a matter of choice; RA 9003 mandates it. At the start, the process may take some learning. But once people begin making proper garbage segregation a habit, over time, it becomes a way of life.

The best way to roll out a solid waste management program in our barangay is to start small and simple. This simple, common-sense guide is designed to show us how to go about it. It has everything we need to start a doable practice of segregation in our household and our community. It will introduce us to widely used techniques. Complicated and complex procedures are translated into easy-to-understand everyday activities. It provides practical tips to make solid waste management less tedious.

Some communities have successfully run a solid waste management program at minimal cost and maximum benefit. All it took really was the willingness to do it and the patience and discipline to get it done. In the end, we will end up with a clean community, with substantial cost savings from hauling waste. By doing so, we will also be doing the environment a big favor.

We invite you to go over the rest of the pages and see how managing our waste can both be simple and easy. It can even be profitable, too.

[Managing our solid waste is no longer a matter of choice; RA 9003 mandates it.]

Why I should participate

EVEN BEFORE we commit to implementing a solid waste management program in our area, we need to be convinced that doing so is worth the community's effort. We must first believe that minding our trash is important. Garbage segregation will become second nature only if it is practiced everyday.

Why do we need to bother with the trash we throw out? What does this have to do with me? These are two basic questions that determine people's willingness and participation in garbage segregation.

Here's a quick answer to these questions: The process it takes for our trash to be disposed – from our household to the garbage truck, then eventually to the dumpsite – costs a lot of money. Each day, each one of us who live in the metropolis generates an average of half a kilo of trash. This means 7,000 tons of garbage need to be hauled to the landfills every day. This is equivalent to 875 garbage trucks that can carry eight tons each.

With this weight, just imagine the billions of pesos of taxpayers' money spent to collect, haul, and dispose our trash. That's just like literally throwing away good money that could have been used more productively, like building more schools and hospitals, hiring more teachers and policemen, and providing more medicines for the poor.

HOW WASTEFUL WE ARE



Of the total solid waste generated from households, a whopping **95%** can still be reused or recycled (45%) or turned into compost (50%).

Only **5%** are made up of residuals (4%) and special/hazardous waste (1%) that no longer have use or not biodegradable.

Waste Analysis and Characterization Survey (WACS), Asian Development Bank, 2003

So what does this have to do with us? A lot! By properly segregating waste at home, in schools, offices, hospitals, malls and restaurants, and other business establishments, we can bring down the volume of trash. The less we dump, the more our local government unit could save from paying private garbage contractors or running landfills. And the more they save, the more public services we get, and the less the need to raise taxes.

By segregating properly, we're also saving a lot of materials that can still be reused, turned into compost for gardens, or even sold as recyclables. This way, we not only help our local government save, we also benefit. Some communities have even started backyard businesses out of recyclables and earn extra income from them.

Less waste means less scavengers swarming over our mountains of garbage. These dumpsites emit harmful gases that make people who live around it sick. Liquid from garbage can seep through the ground and into the ground water table where the community gets the water it drinks. Thus, throwing away our trash is our responsibility to others as well.

As it is, Metro Manila is already running out of areas for dumpsites. The sites we are using right now could last only up to 2010. It's a scary thought. And we have to do something about it. It's time we do our part. Recovering waste for recycling saves our natural resources from being further depleted.



Some communities have even started backyard businesses out of recyclables and earn extra income from them, like this handbag made out of recycled tetrapaks.

Our goals

To make it simple to communicate to people, we can start with four goals:

1. To have clean surroundings;
2. To protect our health;
3. To solve our problem with dumpsites; and
4. To turn waste into useful resources.

[Metro Manila is already running out of areas for dumpsites.]

Defining terms

Here are some of the terms we'll meet when reading this handbook:

Solid waste – also called garbage, trash, rubbish, or junk. It's the stuff we throw from our homes, schools, offices, factories, malls, hospitals, among other things. This includes construction debris, bulky items such as appliances and furniture, waste from the market and farms, and special or hazardous waste from hospitals and factories.

Ecological solid waste management – a way of handling waste to reduce its amount to the barest minimum and recovering discarded materials for reuse, recycling or composting.

Waste segregation – involves sorting garbage or waste into separate containers in our homes so that recyclable items can be sold, waste from the kitchen and the garden can be made into compost, and the remaining garbage (called "residual waste") are sent to the landfill.

Residual waste – what remains after we take out the biodegradable and recyclable wastes. They can't be turned into compost as they're non-biodegradable. Instead, they are hauled off to the dumpsite for disposal.

Dumpsite – a lot where solid waste is brought by dump trucks for disposal. Many people make a living by scavenging recyclable items at the dumpsite.

Sanitary landfill – a more sanitary and better managed disposal facility but more expensive to put up and maintain. A plastic or clay liner separates the waste from the soil to prevent seepage of liquid into groundwater. The dumped waste is spread out by bulldozers, sprayed with deodorizers, covered with soil, and planted with grass.

Leachate (pronounced as *lee-sheyt*) – the waste water that pools under the mountain of garbage. It contaminates groundwater when it seeps into the ground. A landfill has a facility that collects and treats it.

Recycling – recovery of discarded materials such as glass, plastic, steel, aluminum and paper to be used as raw material again. Bottles, cans, newspapers, cardboard packaging and metal are some of the candidates for recycling.

Composting – mixing of kitchen and garden waste with soil to be turned into compost. Microorganisms break down this heap through the usual biological process. When added to soil, compost is used as a conditioner that enriches the soil in gardens and farms.

Biodegradable waste – type of garbage that can be broken down. Includes leftover food, leaves and grass cuttings, fruit, vegetable peelings, bones, egg shell, manure, and the like. This can be turned into compost.

Non-biodegradable waste – type of garbage that cannot be broken down. Includes recyclable materials, residual waste and toxic waste from hospitals and factories.

Republic Act 9003 in a nutshell

WASTE REDUCTION and proper solid waste segregation are no longer choices we have to make. As law-abiding citizens of this country, we are required to follow a law enacted as early as 2001.

Republic Act 9003, or the Ecological Solid Waste Management Act of 2000, was signed into law on January 26, 2001. It is considered a broad-based and comprehensive approach to solid waste management.

It involves segregation at source, segregated collection, storage, transfer, processing, treatment and disposal of solid waste.



UNDP poster on RA 9003 (available in English and Pilipino versions) can be downloaded from this website: www.denr.gov.ph/nswmc/cbeswmp.

The law promotes a way of thinking that waste is a resource that can be recovered. This can be achieved by following the 3 Rs: reduction, reuse, and recycle. The law mandates us to put these principles into practice. By doing so, the problem of solid waste management can be solved.

The law requires the following:

- Solid waste must be reduced at source.
- Recyclable materials must be recovered.
- The remaining waste, after recyclable and biodegradable materials have been separated and used, is to be disposed of properly.

Solid waste management begins inside the household. Starting in our homes, we must learn to conserve resources so we can reduce the amount of materials that we throw away.

Our local government units (LGUs) are responsible for enforcing the law. Government agencies on the national level are required to support LGUs in carrying out this responsibility. LGUs are thus required to draw up a solid waste management plan.

In terms of areas of responsibility, the barangay is tasked to ensure that households and establishments reduce waste, reuse materials, and recover recyclable items. The city or municipality provides a garbage collection system, and proper waste treatment and disposal facility that protect the environment.

GOALS OF RA 9003

- Protect public health and the environment;
- Encourage resource conservation and recovery;
- Promote greater public participation;
- Encourage private sector participation;
- Support research on technologies and techniques in SWM; and
- Promote environmental awareness.



Penalties and sanctions

Under Section 50 of RA 9003, local government officials and officials of government agencies concerned who fail to comply with and enforce rules and regulations shall be charged administratively in accordance with RA 7160 and other existing laws, rules and regulations.

Any citizen may file an appropriate civil, criminal or administrative action in the proper courts/bodies against any person, government entities or official who violates or fails to comply with the provisions of the law. However, the law also protects those who implement it against suits intended to harass or pressure.

Other offenses and penalties under RA 9003 are outlined in Section 48, *Prohibited Acts*, and Section 49, *Fines and Penalties*.

Violations	Penalties
Littering, throwing, dumping of waste matters in public places such as roads, side-walks, canals, esteros and parks, and establishment, or causing or permitting the same. 	*A fine of not less than P300 but not more than P1,000, or rendering of community service for not less than one day to no more than 15 days to an LGU where such prohibited acts are committed, or both.*
Undertaking activities or operating, collecting or transporting equipment in violation of sanitation operation and other requirements or permits set forth in or established pursuant to the Act. Open burning of solid waste.	*A fine of not less than P300 but not more than P1,000 or imprisonment of not less than one day to no more than 15 days, or both.*

[Waste is a resource that can still be recovered.]

Violations	Penalties
Causing or permitting the collection of unsegregated or unsorted waste. Squatting in open dumps and landfills. Open dumping, burying of biodegradable or non-biodegradable materials in flood-prone areas. Unauthorized removal of recyclable material intended for collection by authorized persons.	*A fine of not less than P1,000 but not more than P3,000 or imprisonment of not less than 15 days to no more than six months, or both.*
The mixing of source-separated recyclable material with other solid waste in any vehicle, box, container or receptacle used in solid waste collection and disposal. Establishment or operation of open dumps as enjoined in the Act, or closure of said dumps in violation of Section 37. The manufacture, distribution or use of non-environmentally acceptable packaging materials. Importation of consumer products packaged in non-environmentally acceptable materials.	*A fine of P500,000 plus an amount not less than 5% but not more than 10% of his net annual income during the previous year.* *The additional penalty of imprisonment of a minimum period of one year, but not to exceed three years at the discretion of the court, shall be imposed for the second or subsequent violations of paragraphs 3 and 4 on the right.*



Violations	Penalties
Importation of toxic wastes misrepresented as "recyclable" or "with recyclable content." Transport and dumping in bulk of collected domestic, industrial, commercial and institutional wastes in areas other than centers or facilities prescribed under the Act.	A fine of not less than P10,000 but not more than P200,000 or imprisonment of not less than 30 days but not more than three years, or both.
Site preparation, construction, expansion or operation of waste management facilities without an Environmental Compliance Certificate required pursuant to Presidential Decree No. 1586 and the Act and not conforming with the land use plan of the LGU. The construction of any establishment within 200 meters from open dumps or sanitary landfills. The construction or operation of landfills or any waste disposal facility on any acquifer, groundwater reservoir or watershed area and or any portions thereof.	A fine not less than P100,000 but not more than P1,000,000, or imprisonment not less than one year but not more than six years. If the offense is committed by a corporation, partnership, or other juridical entity duly organized in accordance with law, the chief executive officer, president, general manager, managing partner or such other officer-in-charge shall be liable for the commission of the offense penalized under the Act. If the offender is an alien, he shall, after service of the sentence, be deported without further administrative proceedings.



[Causing or permitting the collection of unsegregated waste is punishable by law.]

Relationship between sectors and the barangay

IN THE scheme of things, the barangay is in the middle of everything. Various sectors play distinctive roles and also have duties to fulfill in solid waste management.

Households are the barangays main "customers." Segregation begins at this level. It is crucial for the barangay to teach households how to segregate their solid waste properly. This also applies to households in depressed areas where narrow roads make garbage collection difficult. Barangays are to collect biodegradable and recyclable wastes.

Commercial, institutional and industrial establishments are the heavy producers of waste. These include restaurants, malls, schools, offices, hospitals, factories and farms. The barangay should monitor the compliance of these sectors with the law. They make good partners in the campaign on waste segregation.

Local government units are responsible for providing a collection system and appropriate waste disposal facility. The barangay should urge the city or municipality to come up with a disposal facility that protects the environment. LGUs are to collect residual waste and provide disposal facilities in accordance with the guidelines set by NSWMC.

Private sector contractors can take care of a barangay's requirements in collection, transfer, treatment and disposal of solid waste. They can even fund and run sanitary landfills, in partnership with the LGU. This sector also includes buyers of recyclable materials such as junkshops and vendors.

Non-government organizations can be a barangay's dependable partner in raising people's awareness on proper solid waste management. It can also provide technical assistance to individuals and groups in running grass roots-level activities that are related to solid waste management.

The National Government and the agencies under it support LGUs in implementing the law. Its main agency on solid waste management is the National Solid Waste Management Commission (NSWMC), an inter-agency body chaired by the Department of Environment and Natural Resources. It was created under RA 9003 to see to it that the law is implemented in every barangay, city and municipality nationwide.

Starting waste segregation



The key: Segregation at source

WHERE DOES the entire solid waste management process begin? Look no further than our home or place of work or study.

The secret to solid waste management is **segregating at source**.

This simply means that as we discard any item or material, we place them in separate containers. It is important that every member of the household, including househelp, drivers, carpenters, gardeners and other workers, understand how trash is segregated. Waste segregation should be practiced at all times.

When practiced in every household and establishment, segregating at source will simplify the barangay's work in managing waste and lead to substantial cost savings.

The four types of waste

1. Recyclables – can be sold to junk shops.
2. Biodegradables – can be turned into compost and used in the garden.
3. Residual Waste – hauled off to the dumpsite.
4. Special or Toxic Waste – also comes from households, hospitals, factories and other establishments but cannot be hauled off to the dumpsites as they require special disposal. In the household, these include motor oil, paints, chemicals, spray cannisters and car

batteries. Motor oil can be brought to gas stations for recycling and car batteries can be traded in at any battery dealer. Medical waste which can be infectious such as syringes, fluids and toxic material require special handling and treatment. Construction debris can be used as filling material.

Caution:
Never mix any of these waste in the same container.

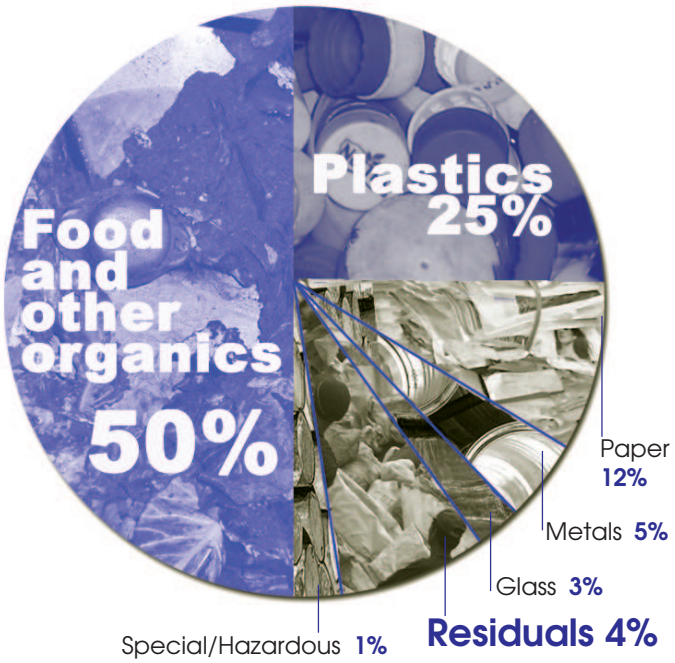




Doing so results in garbage that stinks and attracts flies.

What's in each type

Waste Analysis and Characterization Survey (WACS)
Asian Development Bank, 2003



Recyclables

- Paper – newspaper, folders, used bond paper, corrugated cardboard, *balikbayan* boxes and appliance packaging, boxes used as containers of products such as soap, toothpaste, pizza, gifts etc., wrappers, books, magazines, telephone directory, flyers and brochures, mailing envelopes, laminates (Tetrapak and doy pack packaging).

Tip: Wet paper should first be dried out or can be included in biodegradable wastes. Tin cans can be soaked in soapy water from laundry or bath, then dried. Flatten Tetrapak and paper boxes to reduce space.



- Metal – aluminum cans for softdrinks and beer, tin cans, old roofing sheets, steel gutters, rods and pipes, pots and pans, drums, brass, copper, tire rims and tools.

- Plastic – empty bottles used for juices, soy sauce, vinegar, patis, detergent, shampoo, ice cream containers, plastic monobloc chairs, basins, pails, toys, plastic utensils, egg trays, cups, inkjet cartridges, compact discs and tapes.
- Glass – bottles and jars, broken glasses, plates and windows.
- PET (Polyethylene terephthalate) – bottles from soda and water.
- Old shoes and slippers, clothes, rags, and stuffed toys
- Special household waste such as used motor oil and cooking oil, car or cellphone batteries, and tires.
- Bulky items such as old refrigerators, used computers, airconditioning set, television sets, old washing machines, furniture, beds, and junk vehicles.



THINK ABOUT IT:

We throw away all types of trash but ideally, only **5%** of them should go to the dumpsite.

Recovering 95% of our waste could have saved our government as much as Php3.61 billion a year!

Kitchen Waste

- Food scraps, fish and animal skin and innards, seeds, vegetable and fruit trimmings, coffee grounds, tea bags, used paper napkins
- Shells and seafood
- Egg shells
- Dry animal manure

Garden Waste

- Grass cuttings
- Leaves, twigs and branches



Tip: Remove hard bones and place in residual waste. Bury dead animals and human waste. Dry animal manure before mixing in compost pile. Drain kitchen waste before storing in a covered can or plastic container.

Residual Waste

- Dirty plastic bags
- Used tissue paper, napkins, diapers and cotton
- Foil wrappers and soiled polystyrene or Styrofoam brand packaging used in fastfood packs and coffee cups
- Disposable batteries, pens, cotton swabs, and spray cans



Containers

- Recyclables – place in boxes, sacks or plastic bags. These containers are also recyclable themselves. Content should be dry.

- Kitchen Waste – drain and place in a covered container. A can or a covered pail will do. Never place in a plastic bag.

- Garden Waste – place in a plastic bag. Tie twigs and branches in a bundle for easy handling.

- Residual Waste – place in an old plastic bag. Don't place in a box or sack.

TIP: Don't bring out paper for collection when it's raining so it won't get wet.
A container with a flat base and a secure lid is good for kitchen waste so it can't be toppled over by animals or visited by flies and cockroaches.



Recycling made easy

ONCE PROPERLY sorted out, recyclables can be sold to the nearest junkshop or processing dealers. Arrangements can be made to have the recyclables picked up. Prepare the materials to be sold by putting bottles in sturdy containers like boxes or sacks.

Plastic bottles and aluminum cans can be bundled and placed in plastic bags. Tie flattened cardboard boxes and newspaper piles for easy handling.



Recyclable materials sorted out at source sell higher than when they are recovered from the dumpsite by scavengers.
An obvious reason is segregated recyclables at source are "cleaner" and are not yet stained by garbage. At the Rodriguez landfill, scavenged tin cans sell for P4 per kilo compared with P5.50 to P6 per kilo outside.

I Case Study

Barangay COMEMBO, Makati City:
Adversity breeds opportunity

PROBLEMS with a dumpsite forced a small barangay in Makati City to embrace waste segregation in 1998.

Purificacion D. A. Gonzales, then newly elected barangay captain of COMEMBO (acronym for Combat Enlisted Men's Barrio), was just warming her seat when the San Mateo dumpsite in Rizal closed its doors to trash coming from Metro Manila.

Using an abandoned building as a makeshift MRF, the barangay embarked on a program that was borne out of necessity rather than compliance with the law. At that time, RA 9003 was still a draft bill in Congress. To make compost, the barangay scrounged for funds to buy a composter drum and a shredder. Money eventually came from the 20% development fund in its 1999 budget.

Its plan was to drastically bring down the volume of waste collected from households and the market. Enforcers implemented strict "no segregation, no collection" rules on biodegradable and non-biodegradable wastes. The result surprised even the barangay captain herself. Before garbage segregation, the barangay needed more than one garbage truck per street to haul all the trash. Nowadays, one truck is more than enough. "Political will is very important," says Barangay Captain Gonzales.

COMEMBO now reports 98% compliance with RA 9003. It went on to win numerous awards in Makati City, including the coveted "Cleanest and Greenest Award."

When the dengue epidemic hit Metro Manila, COMEMBO's clean surroundings spared its residents – proof that the barangay's persistence had paid off.



Bgy. COMEMBO's MRF, courtesy of UNDP.

3 / The three Rs: Reduce, Reuse & Recycle

TO BRING down the amount of trash we throw out, experts came out with the three Rs, which stands for Reduce, Reuse and Recycle.

It is possible to limit the trash we generate by simply avoiding the ways we make them in the first place.

Reduce means bringing down the amount of trash we dispose of by consciously avoiding buying items that generate a lot of trash.

For example, an item that comes with heavy packaging or too much wrapping results in unnecessary trash. Disposable items like razors, ballpens, printer ink cartridges also produce waste. Go for refillables. Don't buy poorly made products that don't last or could no longer be repaired. They quickly end up in the dumpsite.

RA 9003 sums it up:

- Avoidance – buy only things you need and avoid items with heavy packaging.
- Product reuse – find creative ways to reuse things and have broken items fixed.
- Increase product durability – go for tough, durable and well-made items.
- Reduce materials used in production – efficiency in manufacturing is called for.
- Decreased consumption – less means less waste.



Reuse items whenever possible. Glass jars can be kept and reused for storing food. Plastic bags can be used a second time. Use cans for flower pots, turn bottles into ornaments, and reuse bond paper for scratch paper.

Recycling recovers materials that can be used as raw materials. It takes only a fraction of the energy and cost to make a bottle from recycled glass than to make one from raw sand.

Practicing the three Rs should bring down the amount of trash we make.

/ Case Study

Barangay Sto. Niño, Parañaque City:
Out of space yet out of waste

FOR a barangay of 40,000 people, Sto. Niño in Parañaque City lacked an open area for an MRF. This limitation, however, did not stop the barangay from pursuing a solid waste management program.

"When UNDP partnered with our barangay in 2003, we had to look for space to put up an MRF," says Barangay Captain Ismael de Leon. "We found a piece of property that's owned by the government."

That piece of property is a strip of space on a sidewalk. The location is unique as it is located on the roadside between sprawling cargo warehouses right across the Ninoy Aquino International Airport. No one would suspect that in this concrete jungle, composting takes place every day.

The shredder and mixer used for composting were donated by the city and UNDP. Mr. Ismael is proud to say 75% of the barangay's 6,600 households now segregate waste. The recyclable materials collected from homes are sold directly to buyers. The MRF is run clean and efficient. Since the MRF was established, the barangay never received a single complaint from its neighbors.

No one would suspect that in this concrete jungle, composting takes place every day.



The benefits of composting

THE MAIN purpose of composting is to reduce the amount of solid waste. Half of the trash we generate are biodegradable. This means it can be broken down by microorganisms in the soil and turned into compost.

Composting our biodegradable waste is highly encouraged under the law as it cuts by half the trash that go into our dumpsites.

Composting is like recycling because we turn organic waste into soil conditioner and even fertilizer for our plants. It enriches, restores and replenishes the topsoil with valuable nutrients. Through composting, organic waste is made useful because it goes back to the cycle as plant food and on into the food chain.

Finished compost is a natural fertilizer that is more environmentally friendly than chemical fertilizers. With the proper volume, finished compost can be sold to farms and homes to enrich garden soil.

What can be composted

Below are some of the items that can be turned into compost:

Kitchen waste
food scraps, fish and animal skin and innards, seeds, vegetable and fruit trimmings, shells and seafood, coffee grounds, tea bags, used paper napkins, egg shells and dried pet or animal waste.

Garden waste
grass cuttings, leaves, twigs and branches, and other woody material.



TIP: For hygienic reasons, human waste and dead animals should not be included in composting. They also cause unpleasant odor. Do not mix in charcoal ashes in the compost as it will kill the microorganisms.

How it's done

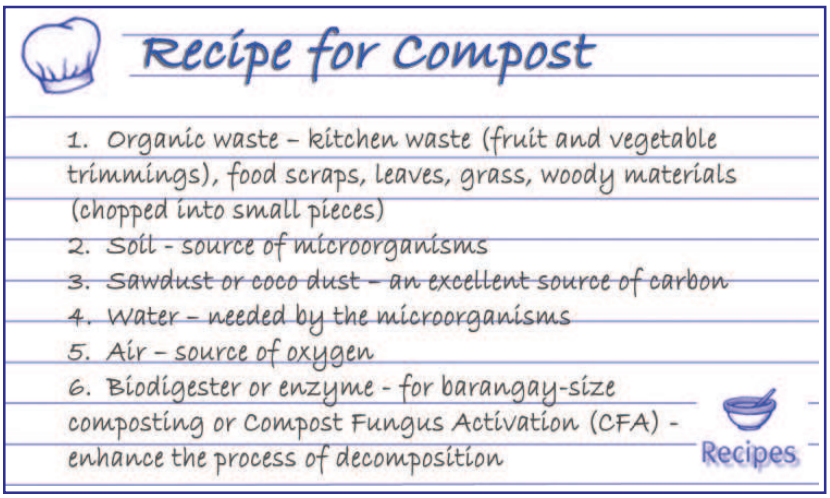
COMPOSTING is easy. It's like baking a cake as we bring together "ingredients" to turn into compost. During composting, microorganisms eat the organic waste and break it down into its simplest parts. When properly done, composting can take two to three weeks to complete. The end-product is a rich, soil-like material that contains carbon and fiber.

The microorganisms need oxygen and warmth to do their work. That's why it's important for the compost to get in air during the process. Microorganisms also need water to live and multiply. That's why we need to water the compost to keep it moist. You will also notice that the compost produces heat during the process. This is normal as the microorganisms give off carbon dioxide and heat as they break down the materials.

Some basic steps

1. Choose a site for the compost pile.
2. Choose a structure for composting.
3. Make the compost by using the ingredients.
4. Attend to the compost pile.
5. Collect and use the compost.

To make compost, you will need the following:



A quick method of making compost
(medium scale)

1. Prepare the ingredients. The ratio is: 50% kitchen waste, 30% garden waste, and 20% coco dust.
2. Mix the ingredients in a heap with a shovel for 5 to 10 minutes. Drain the kitchen waste before mixing. Remove plastic, hard bones and metallic objects. Hard bones will break shredder blades. Food in plastic will not break down. A rotary mixer (used to mix cement) can also be used.
3. To speed up composting, a biodigester or enzyme must be sprayed on the heap while being mixed. The enzyme contains effective microorganisms and can be bought in liquid form from dealers. Mix it with water at a ratio of 1:1 before applying on the mixture. Slowly mix in the enzyme for even distribution. A catalyst containing enzyme in powder form is also available. If this is used, make sure it is mixed evenly.
4. Get a sack and stuff the mixed material inside. Make holes in the sack to let air in. Do not use sacks with plastic lining.
5. Store the sacks in a single layer in a place protected from rain and where air can circulate. Do not stack the sacks at this stage. If stacked high, the contents of the sacks in the lower layers will rot.
6. After a week, the sack will feel warm to the touch. It will also become lighter. If the sack stinks, it means the microorganisms died. When this happens, the compost will not dry up. Repeat the steps when this happens.
7. The compost in the sack will dry up in two weeks. You can pile the sacks when dry but not more than three layers high.
8. In this method, do not water the sacks as it will wash away the biodigester or enzyme.
9. After 25 days, the compost is ready. It should appear humus-like with a sweet earthy smell. If you have a shredder, shred the contents of the sack.
10. Let the shredded compost pass through a screen to come up with fine compost.
11. Mix the compost with soil in the garden as a conditioner or pack in bags for sale.

Starting your own composting

TO START composting in the barangay or your backyard, you will need a compost pile or heap. There are other ways of making compost other than using a sack. Choose a method that is right for the volume of the compost and the availability of space or materials. Below are some of the commonly used methods:



A composter does its work

The heap is the simplest form of composting. Simply pile the ingredients and turn the material with a shovel once in a while for it to get air. The heap should also be watered to keep it moist so microorganisms can thrive. Composting happens slower with this method.

The twin pits requires ample open ground. Dig two square pits one meter on each side and one meter deep. In the middle of the pit, vertically plant a PVC pipe where holes have been drilled on its entire length. The holes allow air in. The pipe can be 3.5 meters long and 4 inches in diameter. Dump the ingredients of the compost on the first pit. There's no need to turn it as it would be difficult since it is below the ground. The perforated pipe will provide continuous supply of air. Transfer the finished compost to the second pit.

The compost bin is simply a box structure placed above ground. The sides can be made of interlaying wood or wooden pallets. This allows air to come in on all sides. Compost is easier to turn once in a while if one of its sides can be opened. Cover the bin to prevent excess rainwater and animals from getting in. Collect the finished compost from the bottom.

The chicken wire is a structure that basically holds together the compost heap above ground. It provides excellent ventilation on all sides. Cover the wire structure to prevent excess rainwater and animals from getting in. Collect the finished compost from the bottom.

The tower tires can be an inexpensive way of composting in the backyard. Just stack old tires and fill in with material. Place rocks or wood in

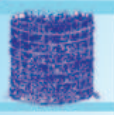
between the tires as spacers for air to get in. Cover the tires to prevent excess rainwater and animals from getting in. Collect the finished compost from the bottom.

Clay flower pots compost uses flower pots. Line up 10 flower pots and fill up with compostable material one by one. As the last pot is filled up in a month's time, the first pot is ready for use as soil conditioner.



TIP: Water the pile or the pit to keep it moist but not soaked. Cover it with plenty of soil, and add new material to the top along with fresh soil.

Various Composting Methods

Heap		*Pros:* Inexpensive, low maintenance *Con:* Pile tends to spread out
Twin pits		*Pros:* Inexpensive, large capacity *Cons:* Pallets can be broken
Bin		*Pros:* Inexpensive, large capacity *Cons:* Slow composting
Chicken wire		*Pros:* Inexpensive, faster composting than bin *Cons:* Labor intensive
Tower tires		*Pros:* Inexpensive, makes use of old tires, unlimited holding capacity
Clay flower pots		*Pros:* Faster composting *Cons:* Pots cost more

I Case Study

Barangay Rosario, Pasig City:
Keeping it simple

HOW do you start a solid waste management program in the barangay?

Begin by deciding to take the program seriously, says Paul Jeffrey Santos of Barangay Rosario in Pasig City. The young barangay captain says replicating his community's experience in Barangay Rosario is easy.

In 2005, the barangay started with a survey on what needs to be done to implement garbage segregation properly. He coordinated with various sectors in his barangay before coming out with an ordinance on solid waste management.

An information campaign was soon launched, teaching people how to segregate waste. They distributed leaflets to all households. The barangay also put up its own MRF where recyclable materials are brought.

"I met with six to seven junkshops and made arrangements with them to buy the recyclable materials," he says.

With UNDP's support, Barangay Rosario received new eco-bikes and a Mobile MRF, the first of its kind in the Philippines. A personal computer was also provided to monitor the barangay's collection of recyclables.



With UNDP's support, Barangay Rosario received new eco-bikes and an MRF to sustain the solid waste management program.

Where to begin

STARTING a solid waste management program in the barangay need not be complicated. Here's a quick checklist of what needs to be done:

1. Create a barangay solid waste management committee where various sectors are represented. Similar boards for the provinces, city and municipality.
2. Draw up a solid waste management plan.
3. Appoint a person to oversee operations and compliance.
4. Pass an ordinance supporting the plan.
5. Announce a collection schedule, in coordination with the LGU.
6. Start a communications campaign informing the community of segregation and waste pick-up schedules.
7. Set up a materials recovery system (MRS) for recyclable material.
8. Link up with junk dealers and recyclers for the purchase of recyclables.
9. Set up a composting facility or push for backyard composting.
10. Monitor results, enforcement and compliance.

To assist you in implementing the checklist, following is a brief description of what each of the items entails.

The barangay solid waste management committee is chaired by the barangay captain. For broader representation, members may include the following:

- One barangay councilor (*kagawad*);
- Head of the youth group (*Sangguniang Kabataan* chair);
- Presidents of homeowners associations;
- Public and private school principals;
- A representative each from the parents and teachers association, religious organization, business sector, non-government organization, market vendors association and the junkshop owners group.

Once formed, the Committee will come up with a program on solid waste management consistent with the plan of the city or municipality. It will also make sure the program is implemented and sustained, and that funds are secured to run it.

Funding may come from the barangay's development fund which accounts for 20% of its annual budget. Other sources of funds are penalties, service fees, or the local solid waste management fund.

Appoint a head to oversee the implementation of the program on a day-to-day basis. The head will regularly report progress to the committee.

An ordinance supporting the program will give the implementation of the law teeth. The ordinance provides the sanctions and penalties that effectively ensures compliance.

A collection system is crucial to ensure segregated transportation of waste. Many programs have failed despite the practice of segregation at source because the transportation of waste continued to be mixed. Coordinate with the barangay, city or municipality on the collection schedule for biodegradables, recyclables, and residual wastes, respectively. Another approach is to employ a dump truck with partitioned cargo space for the three types of waste or separate trucks collecting biodegradable, recyclable and residual wastes.

A communications campaign should be started and continuously run. This should include briefings with homeowners associations, schools and business establishments. Posters can be put up and information materials such as flyers can be distributed in the community. The amount of effort should be similar to running an election campaign – it must be creative, thorough and comprehensive. The materials should communicate what segregation at source is, why it's being done, what the collection schedules are, and the penalties and sanctions for non-compliance.



Poster on the garbage collection schedule in Barangay Rosario, Pasig City

A materials recovery system or MRS is a complete system of recovering wastes that can still be of use – from recyclable materials to kitchen waste that can be turned into compost.



Eco-bike collecting recyclables

Linking up with junkshops and recyclers ensures that recyclable materials are sold. This creates livelihood and promotes enterprise in the community. If the barangay is unable to set up an MRF due to lack of a suitable site or limitation in funds, it can commission junkshops to do the collection of recyclable materials for it.

A composting facility is where the biodegradable materials go. It comprises half of the trash output in the community. In a community where households have backyards, the barangay can push for backyard composting. If the barangay finds adequate space, it can also implement its own composting which can provide employment.

Monitoring of enforcement and compliance is the only way to sustain the program. The barangay should include this in the ESWM plan.

Designing your own MRS

A materials recovery system (MRS) is a complete system of recovering wastes that can still be of use – from recyclable materials to kitchen waste that can be turned into compost. Below is an ideal step-by-step guide in setting up an MRS in your barangay, municipality or city:

STEP 1. Identify sources of wastes and conduct a Waste Analysis and Characterization Study (WACS).

Sources, types and amount of wastes shall be identified to determine the scope of collection and appropriate recovery, treatment and processing.

Households and wet markets would have a lot of biodegradables – vegetable, fruit, fish and meat trimmings; fast food courts and quick-service restaurants (QSRs) would have a lot of Styrofoam, plastic utensils, leftover food, etcetera. Wet markets would focus on composting

and, if they went one step further, they can go into gardening using their compost. Fast food centers should see to it that all styropor and plastic are collected for recycling.

STEP 2. Identify potential collectors.

Potential collectors, such as junk buyers, major recyclers, cultured fish producer, piggeries and poultry owners, and fertilizer producers, should be identified and linked to possible sources of wastes that would serve as their raw materials for recycling, feeds and other uses.

Compostable wastes, such as kitchen and garden waste, need not leave the premises of the home or establishment. Minimal amount of this type of waste can be buried in the backyard or processed in drum composters in case of middle- to high-class subdivisions or commercial establishments.

A significant volume of kitchen leftover from households, quick-service restaurants, and other sources can be used as feed for cultured fish production, piggeries and poultries. These could also be collected by producers of organic fertilizer and liquefied petroleum gas (LPG) to produce methane for cooking.

STEP 3. Conceptualize a suitable MRS.

Based on the waste composition from different sources and the presence of potential collectors, a suitable MRS shall be designed. The MRS shall have four major operational components namely: segregation at source; collection and transport; storage; and marketing and sales. Segregation shall be conducted at source while there could be several options for collection, transport, storage and marketing.

STEP 5. Identify the types of MRS.

Below are the different types of MRS that could be adopted by the barangays, LGUs or private sectors:

MRF in a Barangay or a cluster of Barangays: a facility within the barangay or cluster of barangays to serve as storage area for various types of wastes and composting area for collected biodegradable wastes.

City/Municipal MRF: a centralized facility that can be located at a strategic location within the city/municipality where collected wastes could be delivered in a manner that will not cause traffic problems. Collection could be done by LGU-hired collectors or by each barangay.

Junkshop as MRF: for areas without available space and with limited budget for a construction of a new structure as MRF. The barangay/ LGU can enter into partnership arrangement with junkshop operators.

MRF/Drop-Off Center/Buy-Back Center in Malls: segregation and storage bins for different types of recyclable wastes can be installed at strategic locations in malls. An arrangement between manufacturers/ producers of special wastes and mall operators shall be facilitated through a MOA or contract.

Mobile MRF: could be fabricated for areas without available space for the structure. A 20-footer container van can be modified into a recyclable collector vehicle. A regular collection schedule and route shall be designed and disseminated to the community to facilitate maximum recovery and collection.

Agri-Business sectors as collector of biodegradable wastes: collected biodegradable wastes could be composted and converted into organic fertilizer by major producers. It could also be used as feeds (after processing) by piggeries, poultry owners or producers of cultured fish. Methane extraction may be done by LPG producers out of the collected biodegradable wastes.

The five Es

WHAT the barangay needs to do is actually covered by what is called the "5 Es" – engineering, education, enforcement, environmental organization, and equity. The 5 Es are components of Ecological Solid Waste Management.

- Engineering – covers setting up the support systems, such as the 3 Rs and segregation. This also includes the necessary facilities such as bins, weighing scales, MRFs, bags, and three-sectioned receptacles including the collection system.

- Education – covers information dissemination, education and promotion of the program. This includes printed materials such as brochures, flyers, posters, newsletters as well as seminars.

- Enforcement – covers the policy support and guidelines, monitoring, compliance, sanction, rewards and incentives.

- Environmental Organization – covers the program's organization, administration and delegation of roles and responsibilities.

- Equity – covers funds sourcing and generating revenues.

Having an MRF on a shoestring budget

NOT ALL barangays can afford to set up a materials recovery facility (MRF). While some will have the funds, they may also lack the space. But there are other ways to recover recyclable materials without breaking the bank.

Here are some ways that are being practiced by some communities:

- The hauler will collect the segregated waste to be transported in a truck with partitioned cargo space.

- Households will sell recyclables directly to junkshops and compost biodegradables in the backyard. Hauler will collect residual waste or even biodegradable waste.

- Barangay will put up an MRF, sell the recyclables to junkshops, and make their own compost. Households can also do backyard composting. The hauler will collect residual waste.



UNDP designed this mobile MRF for some pilot areas under its Community-based Ecological Solid Waste Management Programme.

How to make a simple MRF

RA 9003 states that the barangay shall be responsible for setting up an MRF to process the recyclable and biodegradable materials it collects. But what is an MRF?

An MRF is simply a site where collected materials can be stored until they are sold or processed for composting. An MRF is not a dumpsite; it should not hold residual waste. It is also not advisable to do the sorting of unsegregated trash at the MRF for environmental and hygienic reasons.

There are no hard and fast rules on what an MRF should look like. It should, however, provide the following facilities:

- A covered area where various recyclable materials can be separately stored to facilitate sale to buyers.
- An area where composting can be performed.
- Adequate drainage and security.

Since an MRF is run efficiently like a plant, it should always be maintained clean and odor-free. No flies, rats and animals should be present. It should not release leachate from the collected biodegradable waste.

Building a basic MRF is like building any storage space. Here's how:

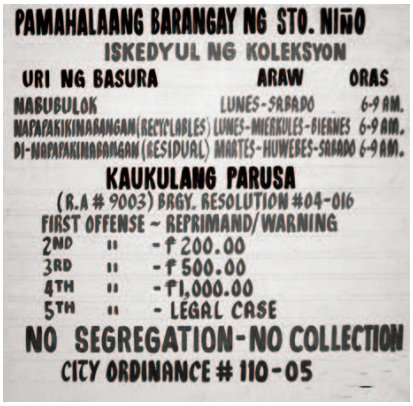
1. Choose a site as small as 12 square meters that is accessible to the barangay.
2. Build a storage space with roofing and partitioned areas protected by cyclone wire. These partitions will hold recyclable materials in separate cells for plastics, glass, metal, aluminum, paper, etc. Hold the area under lock and key to prevent pilferage or theft of the materials.
3. For sites with larger land area, a composting facility can be put up. Construct a mixing pond from concrete where kitchen and garden waste and coco dust can be mixed. The mixing pond should have adequate drainage for the leachate from the kitchen waste and a roofed area for storage of compost sacks. The roofed area will also hold the shredder and mixer.
4. If the barangay chooses other ways of composting, the site should have open ground for compost pits or piles.

Collection and scheduling

Homeowners associations have been successful in implementing collection schedules for segregated waste. Part of the education campaign is to teach people not only to sort out waste but also bring them out for collection on specified days.

If the barangay is able to arrange for segregated collection, it is advisable to assign specific days for the collection of each type of waste. Below is a typical schedule for collection:

- Mondays and Thursdays – residual waste
- Tuesdays and Fridays – recyclable waste
- Wednesdays and Saturdays – garden and yard waste
- Everyday – kitchen waste (which can be picked up by hog raisers directly from the household or the MRF or composted)



Garbage collection schedule in Barangay Sto. Niño, Parañaque City

How to handle waste

Collectors and other personnel directly dealing with collection of solid waste should be equipped with protective equipment. If waste is already segregated at source, collectors are not exposed as much to harmful substances. However, eco-aides doing composting should wear gloves, masks, long-sleeved shirts, pants and shoes to protect them from leachate and dust.

Handling special waste from hospitals and factories follows a different procedure. For purposes of residential, commercial and institutional waste, the basic safety and health equipment should suffice.

Techniques and technology in solid waste management

Biodegradables

Composting

As discussed in Chapter 4, composting is an inexpensive way of reducing the volume of trash we make. It makes use of the natural process of decay and breakdown of organic matter through the action of microorganisms in the soil. To hasten composting, effective microorganisms in enzymes are also used if soil is not present.

Any receptacle will do for composting. It can be a drum, a pail, a stack of old tires, a pit, or a pile above ground fenced off by wood, wire, plastic or none at all. What's important is the organic waste that goes into it and taking care to keep it moist and turned for air. With the right conditions, the microorganisms will do their stuff.

Vermiculture

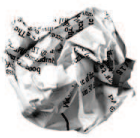
This is an exciting and profitable way of breaking down biodegradable waste using certain types of worms. With the proper structure, these special types of worms, such as the redworm, the African nightcrawler, and the European crawler, can eat through kitchen waste and shredded paper and leaves while producing waste that can later be used as fertilizer. These worms also multiply fast and can be sold at a hefty price.

Biodigester/Bio-reactor

This is an industrial system that makes use of natural processes to treat waste and produce soil improving material, among other things. Biodigesters have been around for some time now and are commonly used for sewage treatment or for managing animal waste. However, as environmental pressures on waste disposal increase, so is the use of biodigesters for reducing waste volumes and generating useful byproducts. It is a fairly simple process that can greatly reduce the amount of waste that could have otherwise ended up in landfills or dumpsites.

Recyclable waste

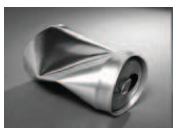
RECYCLABLE materials go back to factories to be turned into products of the same or slightly different form. Recycling saves energy, water, and trees. Below is a table showing how much can be saved through recycling:

When you use:	***What you use to produce***	***What you use when you recycle***
1 ton of paper 	17 mature trees 24,000 gallons of water 28 million Btu energy 1,255 lbs. CO2 released	0 trees 10,080 gallons 19.6 million Btu 890 lbs.
	4.5 tons Bauxite used 1,000 lbs. petroleum coke burned 200 million Btu energy 2,900 lbs. CO2 released	0 Bauxite 0 lbs. burned 9.85 million Btu 145 lbs.
1 ton of glass 	1,330 lbs. sand used 15.2 million Btu energy 441 lbs. CO2 released	0 sand 2.3 million Btu 1.03 pounds
1 ton of plastic 	685 gallons of oil 5,774 KwH energy 98 million Btu of energy	0 gallon 693 KwH 11.76 million Btu

Technologies to Process Recyclable Waste

There are now various companies that offer access to different equipment, processes or technologies to help communities sustain an effective waste management system. Below are some of these alternative technologies:

Types of Recyclable Waste	***Recycling Issues***	***Recycling Process***
Packaging laminates 	• Exhibits superior barrier properties against moisture and oxygen preserving the product in the pack. • Most commonly used as packaging material in budget packs for powdered products	1. Grinding/ crushing 2. Batching 3. Molding/ forming 
Styropor 	• Increased demand resulted in increased volume of waste disposed in dumpsites • Non-biodegradable • About 200 tons of waste styropor generated per month in the country– 70% of which, or 140 tons per month, is in Metro Manila.	1. Melting 2. Molding Finished products: • bricks • blocks • synthetic planks • catwalk tiles 
Glass 	• Should have color segregation to produce predictable color of products. • Heavy investment required but recycled products have export potential. • Recycling extends furnace life and reduces energy costs.	1. Melting 2. Forming 3. Annealing Finished products: 

Types of Recyclable Waste	*Recycling Issues*	*Recycling Process*
Paper and paperboard 	• The Philippines has the second-lowest paper recovery rate in the Asia Pacific Region. • 17% of our waste is paper • 60% energy savings/ 90% less air pollution •Recycling paper also means that there is a substitute available for virgin paper.	1. Segregating 2. Processing of scrap Finished products: raw materials for paper and newsprint, for corrugating medium and test liner 
Plastic bottles 	• The Philippines has the second-lowest paper recovery rate in the Asia Pacific Region. •Recycling paper also means that there is a substitute available for virgin paper.	1. Segregating 2. Processing of scrap Finished products: raw materials for paper and newsprint, for corrugating medium and test liner
Aluminum cans 	• Cost of recycling equipment too high for local investors. • China is to date the largest importer of aluminum scrap from the Philippines.	1. Compacting 2. Reprocessing into raw materials for various products or exported to foreign smelters/ producers Finished products: turned into raw materials
Used tires 		• Holcim uses used tires as Tire Derived Fuel (TDF) in its cement kilns. Finished products: Holcim turns used tires into raw materials for making cement. Some NGOs create pots and sandals from used tires.

For a listing of recycling companies, log on to www.denr.gov.ph/nswmc/cbeswmp.

Residual waste

RESIDUAL WASTE ultimately ends up in the dumpsite or landfill. These are composed mostly of soiled plastic bags, disposal plastic items, foil wrappers, disposable batteries, cloth, film, styropor and other things.

But there are available technologies that can turn residual waste into an additive which is then mixed with concrete binders to make hollow blocks and tiles. An example is Hydromex technology that treats waste with chemicals, shreds it and prepares it for mixing with concrete.

There is also a technology that enables styropor to be turned into tiles by melting it. In India, plastic bags are retrieved, washed and melted together to form a "cloth" which are then made into bags and sold in the United Kingdom. In the United States, there are already disposable pens, utensils and bags that are biodegradable as they are made from cornstarch.

Toxic waste

Medical waste from hospitals are treated in special facilities. Hospitals generate 9,855 tons of infectious waste annually, or 27 tons a day. *(Source: ADB)*

However, certain household chemicals require special disposal. Used motor oil can be returned to gas stations for disposal. Used oil can still be made as fuel for large engines. Car batteries are recyclable and must be returned or traded at the dealers.



Source: www.epa.gov

Did you know that...



- Paper accounts for 12% of Metro Manila's solid waste that goes to dumpsites. This means 840 tons of paper are thrown away daily – equivalent to 14,280 trees.
- Plastics, which account for 25% of Metro Manila's solid waste, require 100 to 400 years to break down at the landfill.
- The energy saved from recycling one glass bottle is enough to run a 100-watt light bulb for four hours.

/ Case Study

Municipality of Navotas

More stick than carrot

MAYOR Tobias Reynald "Toby" M. Tiangco hates litterbugs. In fact, a sure way of making his blood boil is the sight of unswept streets in his part of Metro Manila. There is no excuse for dirty streets in Navotas as there are 500 street sweepers – one for every block – who is also a resident in the area.

Navotas is considered the catch basin of trash coming from Metro Manila as it is where most rivers and esteros end. Trash from all over the metropolis find their way to Navotas before it exits to Manila Bay. Fishing out the trash is a never-ending expense for the municipality.

About a fourth of Navotas' population live in depressed areas. This comprises about 70,000 of the municipality's 250,000 population. It's tough to convince people in depressed areas to segregate because trucks can't squeeze into narrow roads, says the mayor. But Barangay Tanza in the municipality seems to be making inroads. In this barangay, discipline is a sacred word. An NGO works with the barangay to ensure a working solid waste management program.

Mayor Tiangco says he leads with an iron fist. He compels all 40 barangays to allocate their development fund (20% of their budget) to build MRFs. Navotas also swarms with environmental police who issue violation receipts whenever an ordinance is broken. Violators are fined P300 or made to serve the community for eight hours.



For any solid waste management program to be successful, discipline is key, says the 38 year-old mayor.

How a dumpsite works

MOST OF the solid waste in the Philippines are disposed of in open dumpsites. There are now 794 open dumpsites taking in garbage from all 17 regions nationwide, as against 309 controlled disposal facilities. It is the responsibility of the city or municipal government to put up adequate disposal facilities for their constituents.

Most dumpsites begin with the first load of trash dumped onto it. Over time, this piece of land becomes a mountain of trash.

At a dumpsite, anything goes. The trash does not get covered up with soil and are simply piled up on top of the heap. Methane coming from decaying organic material sparks off spontaneous combustion in many places, creating thick smoke and smog. Leachate penetrates the ground and contaminates the groundwater, rivers and streams. A dumpsite also gives out an overpowering stench that makes our eyes water. It is home to rodents, flies and hundreds of people who have made scavenging their lifeblood.

Metro Manila

Around 7,000 tons of solid waste from Metro Manila's 17 cities and municipalities are hauled off to the dumpsite everyday in 1,675 truckloads. It takes 611,375 truckloads a year to bring the metropolis' 2.5 million tons of trash to about a dozen dumpsites.

Metro Manila spends Php3.8 billion a year just to get rid of its trash. This trash is hardly segregated. Mixed in it are recyclable and biodegradable materials that comprise roughly 95% of the total volume.

It is no secret that recyclable materials are recovered from dumpsites only when scavengers sift through every tipped load of trash and pick them up. Just the same, Metro Manila has already spent billions hauling the trash to the dumpsite when 95% of the materials could have already been recovered inside our homes. With proper waste segregation at source, Metro Manila can save Php3.61 billion from its hauling expense and spend only for hauling the remaining 5% of its waste.

The Rodriguez Sanitary Landfill

A HUGE portion of the collected waste of more than 12 million people of Metro Manila (except Quezon City) ends up in the Rodriguez sanitary landfill in Montalban, Rizal. A private corporation that has partnered with the local government runs this 14-hectare facility nestled between rolling hills and hidden from view.

At a dumping fee of Php600 per ton of trash, it is but natural for the local government and the private firm to welcome Metro Manila's unsegregated trash here. The partnership merely dumps the 3,000 tons from 500 truckloads of trash a day in a valley 120 meters deep and covers it up with soil using bulldozers. A plastic liner halts the leachate from the mound and pipes it up to a collecting pond. The facility runs round-the-clock, 24 hours a day, every day.

Around 1,500 people come in two shifts daily to scour the dump for anything of value – from recyclable PET bottles and cans, to sometimes cash or jewelry mixed with the trash. A scavenger earns up to Php250 a day doing the rounds.



Home to Metro Manila's trash

Garbage disposal work brings in more revenues to this sleepy town of Rizal than any of its industries. Jaime S. Baltazar, administrator of the landfill, says revenues of the municipality have tripled since the landfill first welcomed its first truckload of garbage on January 17, 2002.

At Php600 tipping fee for every ton of trash, the landfill earns as much as Php4,800 per dumptruck or Php2.4 million from the 500 trucks on the average that troop to the site on a typical day. The local government of Rodriguez gets a fourth of the tipping fee as share, while the private operator gets the rest through the Metro Manila Development Authority (MMDA).

The Payatas Controlled Dumpsite

NOW KNOWN as the Quezon City Controlled Disposal Facility, this 24-hectare open dump has seen worse days.

On July 10, 2000, a trash avalanche on a slope of the towering mountain of garbage buried 205 people alive. The tragedy highlighted the dangers of uncontrolled dumping and put the spotlight on the government. This tragedy was instrumental in the enactment of the Ecological Solid Waste Management Act of 2000.

After the tragedy, the dump was ordered closed but was reopened four months later.

Today, Payatas is an icon in controlled disposal. Run by a private contractor, it takes in only trash from 2.5 million people living in Quezon City.



(Top photo) The new Payatas is no longer an eye sore. (Below) Visitors are welcomed with a hot towel at the office of the Payatas Operations Group.

Payatas is composed of two mounds of garbage. The old mound has been covered up with soil and planted green with grass. A network of pipes capture the methane gas rising from the mound. The gas is used to fire up a one-megawatt power generator. It has generated enough power to light up homes in Barangay Payatas since 2002. It is estimated that the mound has enough methane to run for 10 years.

The second mound remains in operation. Everyday, 450 trucks from the city haul in 1,200 metric tons of trash to the mound. This load is composed of unsegregated trash, 68% of which are biodegradable waste. Recyclable waste, comprising 19% of every truckload, is systematically picked up by a group of scavengers numbering some 300 at any one time. As many as 10,000 families still depend on dumpsite operations.

A mere 13% of every truckload is made up of residual waste. In effect, only one out of 10 truckloads of trash brought by Quezon City to Payatas had been segregated at source. Quezon City spends Php40 million a month for hauling trash to Payatas.

Dangers in a dumpsite

IN JULY 2000, a huge mountain of trash in Payatas, Quezon City crashed on people scavenging in the dumpsite. The trash avalanche buried at least 200 people in filthy refuse. The tragedy highlighted the dangers of uncontrolled dumping of solid waste. It also brought to the spotlight the direct impact of the trash we throw out of our homes everyday.

In a dumpsite, trucks haul in mixed trash. The pile lacks a liner such that the waste is in direct contact with the ground. Leachate from the trash seep into the ground and contaminate underground water. Decaying trash let off methane that affects our lungs and starts off spontaneous fires all around. It's normal to see a pall of acrid smoke constantly rising at the dumpsite.



The day the mountain of trash trembled and gave way

Oftentimes, when the pile gets high and the sides of the mountain of trash becomes steep, trash slides occur. When it rains, water runoff flows into streams and pollutes the rivers connected to it. Flies, rats and mosquitoes abound. The stench from the dumpsite also poses grave health hazards to the people living in communities around it.

How a sanitary landfill works

A sanitary landfill is a dumpsite designed to handle trash that it holds. A plastic or clay liner provides a layer between the trash and the ground. This liner has channels that collect leachate, which is then pumped through pipes to a collecting pool for treatment. Trash at the landfill is compacted to save on space. Each layer of trash that is spread out by bulldozers are covered with soil before accepting another layer over.

Landfills feature a planned drainage system, a means of collecting methane gas, fencing, roads, disinfection for the trucks and safety structures. That's why they are expensive to run. When a landfill reaches its capacity, it is planted with grass or trees. They can be useful only as golf courses or parks.

A typical sanitary landfill needs anywhere from P30 million to P50 million per hectare to run. Most small municipalities nationwide cannot afford this investment.

The National Solid Waste Management Commission (NSWMC) estimates that seven out of 10 cities and municipalities could not afford to put up their own sanitary landfills. More so, as many as 1,200 local government units nationwide generate less than 15 tons of garbage a day, making landfills untenable.

Landfills are run by private companies as a business. They are normally contracted by the city or the municipality.

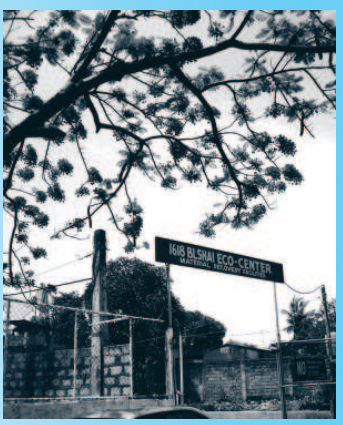


Bacteria in the landfill break down the trash in the absence of oxygen as the landfill is airtight. This produces landfill gas, which contains 50% methane and 50% carbon dioxide. This presents a hazard as the methane can explode and/or burn.

Source: www.howstuffworks.com

1618 Sunflower Green:
The true spirit of community

IN Parañaque City, a small patch of community takes recycling to heart.



Annex 1618 Homeowners Association in Better Living Subdivision has been practising waste segregation at source since March 2001. Volunteer residents, now numbering more than 40, run the 1618 Sunflower Green program in the middle-class community of around 2,000 residents. Some 300 households follow a strict collection schedule where recyclables, residuals, kitchen and garden waste are collected on separate days.

The waste is brought to an MRF the community put up beside its park. At the start, waste was collected using a pedicab and later a second-hand truck. In 2004, 1618 bought a new truck funded by a novel campaign. Called *Patak Patak para sa Eco-truck*, residents chipped in any amount they could spare everyday during the 30-day fundraising campaign. Volunteers knocked on every door and collected coin donations in used PET bottles.



A novel fundraising scheme enabled the 1618 community to buy their own eco-truck.

Today, 1618 sells an average of P15,000 worth of recyclables every month. Through the 1618 Sunflower Green program, the community is able to save around P70,000 a month in private hauling fees. From March 2001 to May 2006, it avoided paying a total of P2.2 million in hauling fees to a private garbage collector. 1618 residents were able to keep their monthly dues at only P300 over the past five years because of the program.

8 / Monitoring and Evaluation

A step-by-step guide

FOR every community-based ecological solid waste management system to work, it is important to follow certain steps in monitoring and evaluating the program. Below is an ideal step-by-step process flow for the municipality/ city and the barangay:

FOR THE CITY/MUNICIPALITY

STEP 1. Initial Waste Appraisal

The LGU and representatives of each barangay should first determine the profile of waste in the city or municipality using a list of buyers and residuals generated. They should identify problems related to social, cultural, political, and economic issues, and find out how each barangay practices solid waste management.

STEP 2. LGU-Citizens Consensus Building

With the help of a workshop facilitator, local leaders and citizens should agree on a vision, strategies, and first steps to address the problems. From this workshop, the LGU must be able to come up with consolidated lists that could be used for compost, recyclables, and residual waste management.

STEP 3. Multi-stakeholder Planning and Organizing

A group composed of various stakeholders, leaders and volunteers should review options; set standards, objectives and targets, action plans, and budgets. This must result in doable actions for managing compost, recyclable, and residual waste. Assign key people with corresponding roles and responsibilities that are time-bound. Come up with a list of recyclable buyers and residuals.

STEP 4. Implementation of Doables

The LGU should carry out designated roles, and enact policies to support program objectives. It should promote the three Rs (recycling, reuse, reduction) through information and education campaign materials. It must also set monitoring standards for managing residual waste and running an

MRF or a storage facility. It could also use a list of buyers (junkshops) for the waste it collects, and keep a list of residuals for waste collection and recordkeeping.

STEP 5. Monitoring of Citizens' Compliance

The LGU and a multi-sectoral community of volunteers conduct appraisals in the barangays. This should determine: the number of barangays complying with RA 9003 based on their list of residuals generated, amount of residual waste (per item) collected in a given time, recycling efforts being made, and waste reduction schemes. Update the lists of residuals, recyclables, and compostable waste generated and have them validated by the community to increase public participation and support.

STEP 6. Solid Waste Management Board Review

Review the results of monitoring data and recommend needed interventions. The entire monitoring process may either end here or return to Step 2, depending on the need of the LGU.

FOR THE BARANGAY

STEP 1. Initial Waste and Community Appraisal

The barangay and community volunteers should first determine the profile of participants and the process of generating recyclable and residual waste (using participatory dialogue). This should be validated by the community on a per *sitio*/*purok* (district) level. From this activity, the barangay must be able to determine the waste profile, volume, existing solid waste management practices (if any), and identify problems related to social, cultural, political, and economic issues.

STEP 2. Barangay Council-Citizens Consensus Building

Consolidate *sitio*/*purok* maps and lists, and use these as one of the bases for local leaders and citizens to agree on a vision, strategies, and steps to address the problem/s.

STEP 3. Multi-stakeholder Planning and Organizing

A group composed of various stakeholders, leaders and volunteers should further review options; set objectives and targets, action plans, and budgets for doable actions. Assign key people with corresponding roles and responsibilities that are time-bound. Come up with a list of recyclable buyers and residuals.

STEP 4. Implementation of Doables

The barangay council should enact policies to support program objectives. It should assign key roles to promote the three Rs (recycling, reuse, reduction). Through the use of information and education campaign materials, the barangay must enlist community participation in segregation at source, segregated collection, and storage.

STEP 5. Monitoring of Citizens' Compliance

The barangay council and a multi-sectoral community of volunteers must update validated *sitio/purok* maps and the recyclable and residual lists generated from Step 1, and conduct appraisals. The appraisal should determine: the number of households complying with RA 9003, the volume of segregated waste collected in a given time, recycling efforts being made, and waste reduction schemes. Update maps and lists and have them validated by the community to increase public participation and support.

STEP 6. Barangay Solid Waste Council Review

Review the results of monitoring data and recommend needed interventions. The entire monitoring process may either end here or return to Step 2, depending on the need of the barangay.

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